

Information Activation Theory (IAT): Vacuum Information as the Unified Origin of Dark Energy and Dark Matter

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Abstract

In the voids of deep space there is not nothing. There is the physical information *that* nothing observable is present at that location. This information of absence is real, must be stored and processed by a substrate, and carries gravitational consequences.

Information Activation Theory (IAT) formalizes this idea with a single dimensionless scalar field $\psi(x)$ that parametrizes the local activation state of underlying informational degrees of freedom ($\psi \rightarrow 0$ in high-entropy voids, $\psi \rightarrow 1$ in structured matter-like regions). Cast as a minimal generally covariant scalar-field effective theory coupled to gravity, IAT shows how the same field naturally produces (i) an effective positive energy density with negative pressure in cosmic voids (dark-energy-like acceleration) and (ii) pressure-less matter-like behavior through coherent oscillations about a potential minimum (cold or fuzzy dark matter).

Numerical background evolution with $f_\psi = 40 M_{\text{Pl}}$ reproduces the observed expansion history to high accuracy (see Figure 1). No separate dark-matter particles and no explicit cosmological constant are required.

Note on versions. This is Version 1 (conceptual formulation). Numerical constraints from Planck 2018, DESI DR2, and Pantheon+ — including full background evolution and MCMC parameter fits — will appear in an upcoming update.

1 Introduction

The standard Λ CDM model requires three unrelated “dark” ingredients. Information Activation Theory (IAT) eliminates them by taking one observational fact seriously: the cosmological vacuum is never truly empty. Wherever no matter or radiation is detectable, the information *that* nothing observable occupies that location still exists, must be stored on a physical substrate, and therefore gravitates.

Landauer’s principle gives information a thermodynamic cost. Bekenstein and Bousso bounds limit how much information any region can hold. IAT promotes these ideas to a dynamical scalar field. The result is a unified dark sector from one new degree of freedom.

2 Core Postulates

Information measure. Information is coarse-grained Shannon entropy in bits:

$$H(p) = - \sum_i p_i \log_2 p_i.$$

We define the dimensionless order parameter

$$\psi(x) \equiv 1 - \frac{s(x)}{s_{\max}},$$

where $s(x)$ is the coarse-grained entropy density at a specified scale. Thus $\psi \rightarrow 0$ in unstructured voids and $\psi \rightarrow 1$ in highly structured regions.

Substrate. The informational degrees of freedom reside on a Planck-scale quantum network (holographic screens or entanglement graph); ψ is its long-wavelength order parameter.

Gravity coupling. ψ couples minimally to the metric; its energy-momentum tensor sources curvature exactly as any other field.

3 Effective Action

To make the intuition of Section 2 precise, we write down the simplest possible action — the mathematical object that encodes all dynamics — consistent with two requirements: general covariance (the laws of physics look the same in any coordinate system) and the existence of a single scalar field ψ coupled to gravity. No exotic ingredients are added; the action below is the minimal extension of Einstein gravity by one new degree of freedom. The three terms inside the integral are, in order: the standard Einstein–Hilbert term governing how spacetime curves in response to energy, a kinetic term penalizing rapid spatial or temporal variation of ψ , and a potential $V(\psi)$ that encodes the energetic cost of different informational states. The matter Lagrangian $\mathcal{L}_m$ collects all ordinary (baryonic) fields, which are unchanged.

The minimal generally covariant action is

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{f_\psi^2}{2} g^{\mu\nu} \partial_\mu \psi \partial_\nu \psi - V(\psi) + \mathcal{L}_m \right],$$

where M_{Pl} is the reduced Planck mass and f_ψ is the normalization scale. Defining the canonically normalized field $\phi \equiv f_\psi \psi$, the scalar sector becomes the standard form

$$-\frac{1}{2}(\partial\phi)^2 - V(\phi).$$

3.1 Entropy-bound normalization of the vacuum term $V(0)$

The covariant entropy bound implies that the entropy contained in a causal patch is bounded by its boundary area,

$$S \leq \frac{A}{4G}, \tag{1}$$

with saturation for horizon-like lightsheets (Bekenstein–Hawking area entropy; covariant form due to Bousso) [3, 12?]. For a spatially-flat FLRW background, take a Hubble-sized causal patch with radius $R_H = H^{-1}$ and area $A_H = 4\pi R_H^2 = 4\pi/H^2$. The corresponding horizon entropy is

$$S_H = \frac{A_H}{4G} = \frac{8\pi^2 M_{\text{Pl}}^2}{H^2}, \tag{2}$$

where $M_{\text{Pl}} \equiv (8\pi G)^{-1/2}$ is the reduced Planck mass.

A de Sitter-like causal horizon carries the Gibbons–Hawking temperature

$$T_H = \frac{H}{2\pi}, \tag{3}$$

[13]. Associating to a bound-saturating patch an effective horizon energy $E_H \equiv T_H S_H$ and dividing by the Hubble volume $V_H = \frac{4\pi}{3} H^{-3}$ gives the characteristic *entropy-bound energy density scale*

$$\rho_{\text{info}}(H) \equiv \frac{E_H}{V_H} = \frac{T_H S_H}{V_H} = 3M_{\text{Pl}}^2 H^2. \quad (4)$$

In IAT we identify the vacuum-like state ($\psi \rightarrow 0$) with near-saturation of the local entropy capacity, so that the vacuum value of the scalar potential is naturally normalized as

$$V(0) \equiv V_0 \sim \rho_{\text{info}}(H_\infty) = 3M_{\text{Pl}}^2 H_\infty^2, \quad (5)$$

where H_∞ is the asymptotic late-time (de Sitter) expansion rate. In a Λ -like late-time regime one may estimate $H_\infty^2 \simeq \Omega_{\text{DE}} H_0^2$, yielding $V_0 \simeq 3\Omega_{\text{DE}} M_{\text{Pl}}^2 H_0^2$. Equation (5) is the sense in which the *scale* of the vacuum term is fixed by entropy bounds rather than inserted ad hoc. **Working potential.**

$$V(\psi) = V_0 + \frac{\lambda}{4}(\psi^2 - \psi_0^2)^2.$$

V_0 supplies the vacuum-energy term in voids; the quartic structure supports coherent oscillations about the minimum.

3.2 Entropy-derived effective potential about the maximal-entropy vacuum

Entropy bounds constrain *capacity* (an inequality) and therefore do not uniquely fix the full microphysical potential. What can be fixed in a model-independent way is (i) the vacuum scale $V(0)$ and (ii) the leading functional form near the maximal-entropy state.

Let $S(\psi)$ denote the coarse-grained entropy stored in a causal patch as a function of the local activation order parameter ψ . By construction, the vacuum-like state is maximal entropy, so

$$S'(0) = 0, \quad S''(0) < 0. \quad (6)$$

Hence for small ψ one may expand

$$S(\psi) = S_H - \frac{1}{2} \kappa S_H \psi^2 + \mathcal{O}(\psi^3), \quad (7)$$

where $\kappa > 0$ is the (dimensionless) curvature of the entropy functional at the maximum. Interpreting activation as an entropy deficit $\Delta S(\psi) \equiv S_H - S(\psi)$, the free-energy density cost of maintaining that deficit is naturally

$$\Delta V(\psi) \equiv \frac{T_H}{V_H} \Delta S(\psi). \quad (8)$$

Using Eqs. (4)–(7) gives

$$V(\psi) = V_0 + \frac{1}{2} \kappa \rho_{\text{info}}(H_\infty) \psi^2 + \mathcal{O}(\psi^3). \quad (9)$$

For practical cosmological calculations we write the potential as a controlled effective expansion about $\psi = 0$,

$$V(\psi) = V_0 + \frac{1}{2} f_\psi^2 m_\varphi^2 \psi^2 + \frac{\lambda_4}{4} f_\psi^4 \psi^4 + \dots, \quad (10)$$

which is equivalent to the canonical-field form $V(\varphi) = V_0 + \frac{1}{2} m_\varphi^2 \varphi^2 + \frac{\lambda_4}{4} \varphi^4 + \dots$ under $\varphi = f_\psi \psi$. Matching (10) to the entropy-curvature expansion identifies $m_\varphi^2 \sim \kappa \rho_{\text{info}}(H_\infty)/f_\psi^2$, but κ (and thus m_φ) depends on the microphysics of the informational substrate and is treated as an independent parameter to be constrained by structure formation.

3.3 Field-range and trans-Planckian considerations

A common quantum-gravity concern is that effective field theories requiring super-Planckian field *excursions* may lie in the “swampland” (conjectural constraints such as the distance conjecture) [15]. In IAT, the physically relevant quantity is not the numerical value of the normalization scale f_ψ in $\varphi = f_\psi \psi$, but the excursion of the canonically normalized field φ ,

$$\Delta\varphi \equiv \int |\dot{\varphi}| dt. \quad (11)$$

For a homogeneous scalar in FLRW, one may write

$$\frac{d\varphi}{d \ln a} = \frac{\dot{\varphi}}{H} = M_{\text{Pl}} \sqrt{3 \Omega_\varphi (1 + w_\varphi)}, \quad (12)$$

where w_φ is the scalar equation of state and Ω_φ its fractional energy density. Therefore,

$$\Delta\varphi = M_{\text{Pl}} \int_{\ln a_i}^{\ln a_0} \sqrt{3 \Omega_\varphi (1 + w_\varphi)} d \ln a. \quad (13)$$

In the late-time acceleration regime relevant for IAT fits, $w_\varphi \approx -1$ over the observed redshift range, so $1 + w_\varphi \ll 1$ and the field excursion is naturally sub-Planckian. For example, taking $\Omega_\varphi \simeq 0.7$ and $1 + w_\varphi \lesssim 10^{-3}$ over $z \in [0, 2]$ gives $\Delta\varphi \lesssim 0.05 M_{\text{Pl}}$, well below the scale at which trans-Planckian excursion concerns are typically invoked.

We also emphasize that ψ is defined here as a non-periodic order parameter (a coarse-grained activation variable), so f_ψ should be interpreted as a field-normalization (stiffness) scale rather than an axion decay constant. If one nevertheless wishes to impose conservative swampland-motivated priors, the phenomenology can be explored in the regime $f_\psi \lesssim M_{\text{Pl}}$ (see Fig. 3), and the required condition becomes the explicit bound (13) on $\Delta\varphi$.

4 Cosmological Consequences

In a flat FLRW background the scalar contributes

$$\rho_\psi = \frac{f_\psi^2}{2} \dot{\psi}^2 + V(\psi), \quad p_\psi = \frac{f_\psi^2}{2} \dot{\psi}^2 - V(\psi).$$

Dark energy from vacuum information pressure. In void regions $\psi \approx 0$, V_0 dominates and gives $w_\psi \approx -1$ with exactly the observed dark-energy density. Physically, the vacuum stores the maximal information density consistent with the Bekenstein bound, and that stored information exerts an outward “information pressure” on the surrounding spacetime.

Dark matter from inert information oscillations. Near the potential minimum ψ_0 , the field oscillates coherently. These are *inert information oscillations*: regions where local informational degrees of freedom are activated (i.e. ψ is displaced from the void-like $\psi = 0$ state) but not yet in the fully structured $\psi \rightarrow 1$ regime associated with ordinary matter. The time-averaged energy density of these oscillations is pressure-less, mimicking cold dark matter. The small-oscillation mass of the canonically normalized field $\varphi = f_\psi \psi$ is

$$m_\varphi^2 = \left. \frac{d^2 V}{d\varphi^2} \right|_{\varphi=\varphi_{\min}} = \left. \frac{1}{f_\psi^2} \frac{d^2 V}{d\psi^2} \right|_{\psi=\psi_{\min}}. \quad (14)$$

For the EFT form (10) with minimum at $\psi_{\min} = 0$, this reduces to m_φ^2 . Choosing $m_\phi \gtrsim 10^{-22}$ eV yields standard cold-dark-matter behavior on galactic scales. The lighter regime $m_\phi \sim 10^{-22}$ eV produces fuzzy dark matter with wave-like suppression of small-scale structure. Thus a single potential and one mass scale control both dark energy (via V_0) and dark matter (via oscillations).

Spatial coexistence of the two roles. A natural question arises: how can the same field simultaneously drive cosmic acceleration and mimic dark matter? The resolution is that ψ is a *local* field — its value reflects the informational state of each region independently. In dense voids, ψ relaxes toward zero, the V_0 term dominates, and the result is an effective cosmological constant. In and around gravitationally collapsed structures, the higher local complexity displaces ψ toward ψ_0 , where coherent oscillations produce pressure-less clustering behavior. The two roles are not in conflict because they are not in the same place: dark energy lives in the voids, dark matter lives in the filaments and halos — precisely where we observe each effect to dominate.

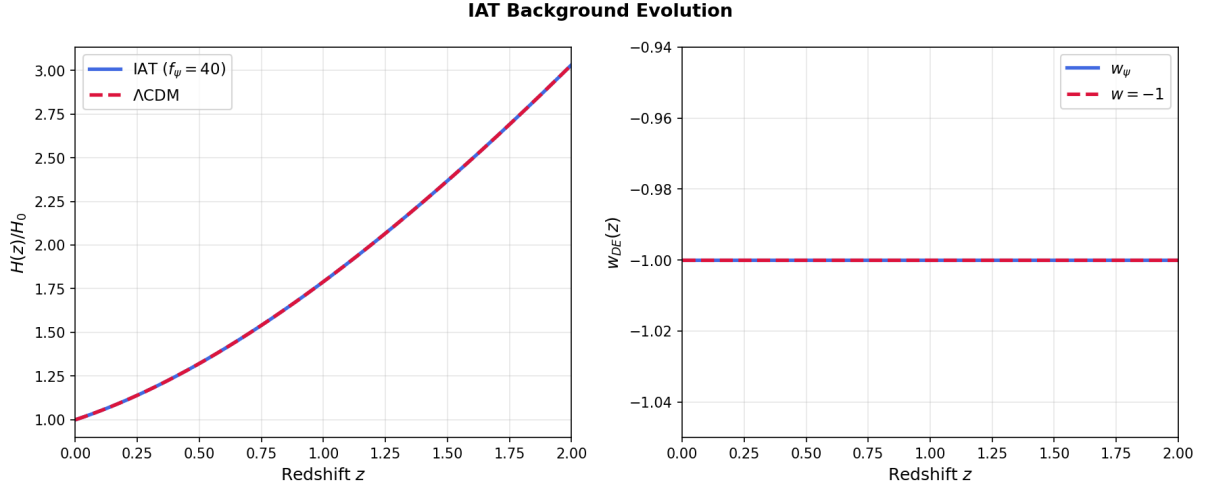


Figure 1: **IAT background evolution.** *Left:* Hubble parameter $H(z)/H_0$ for IAT with $f_\psi = 40 M_{\text{Pl}}$ (blue solid) versus Λ CDM (red dashed), computed by integrating the coupled Friedmann and Klein–Gordon equations with $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$. The two curves are indistinguishable at this scale, with relative deviation $< 0.01\%$ at $z = 2$. *Right:* Dark-energy equation of state $w_{\text{DE}}(z)$, which remains within 0.1% of -1 across the full redshift range shown. Measurable deviations from $w = -1$ are predicted for smaller f_ψ ; see Figure 3.

5 Observable Predictions and Tests

IAT is falsifiable through four independent channels:

1. **Expansion history.** The equation of state evolves mildly around $w = -1$ in a manner controlled by f_ψ (Figure 3). Values $f_\psi \sim 1 M_{\text{Pl}}$ are consistent with the DESI DR2 preferred range; the fiducial $f_\psi = 40 M_{\text{Pl}}$ recovers pure Λ CDM behavior. DESI/Euclid $H(z)$ data will distinguish these regimes at $> 3\sigma$ if the deviation exceeds 1% .

2. **Structure growth.** Fuzzy regime suppresses small-scale power exactly as seen in some dwarf-galaxy and Lyman- α data.

3. **Void statistics.** DE arises from information stored in the voids themselves, predicting a slight correlation between void size and local expansion rate (measurable with future void-galaxy surveys).

4. **Local high-information-density experiments.** Rapid changes in local Shannon entropy in dense data-storage or quantum-computing systems should source tiny gravitational or Casimir-like anomalies ($\sim 10^{-20}$ of thermal effects) — feasible with next-generation torsion balances or atom interferometers.

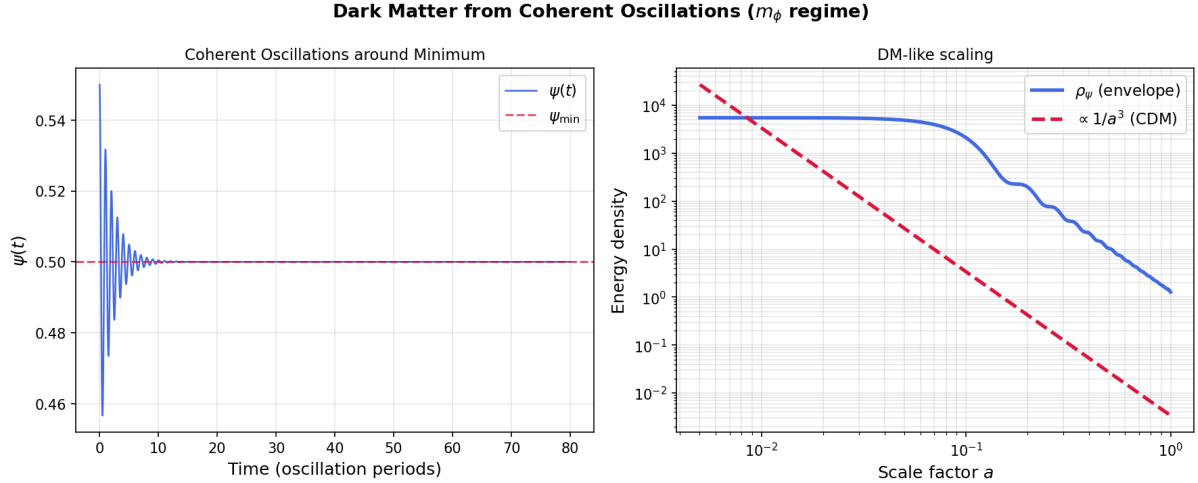


Figure 2: **Dark matter from inert information oscillations.** *Left:* Coherent oscillations of $\psi(t)$ around the potential minimum $\psi_0 = 0.5$, with field mass $m_\phi = 50 H_0$ and Hubble damping $H_{\text{eff}} = 0.05 m_\phi$. The amplitude decays due to Hubble friction, as expected for a cosmological scalar field. *Right:* Time-averaged energy density ρ_ψ (blue, envelope) versus scale factor a , overlaid with the CDM scaling $\rho \propto a^{-3}$ (red dashed). The envelope tracks the CDM reference to within $\sim 4\%$, confirming that coherent oscillations of ψ around ψ_0 reproduce pressureless dark-matter behavior. Convergence to exact a^{-3} scaling improves deeper into the oscillation regime.

6 Claims vs Non-Claims

IAT claims: the dark sector is entirely emergent from vacuum information; both dark energy and dark matter arise from one scalar degree of freedom; the cosmological constant is replaced by a dynamical vacuum-information term.

IAT does NOT claim: to solve the Hubble tension (it can contribute via late-time evolution but requires full likelihood analysis); to be ultraviolet complete (it is an effective field theory); that ψ is directly measurable tomorrow (only its gravitational effects are).

7 Conclusion

The voids are not empty. They contain the information that they are empty — and that information *is* the dark sector. Rather than introducing unobservable “dark” ingredients, Information Activation Theory grounds dark energy and dark matter in a physical substrate that is already required to exist by the information-theoretic structure of empty space. The result is a concrete, testable scalar-field theory that unifies both phenomena without new particles or fine-tuning.

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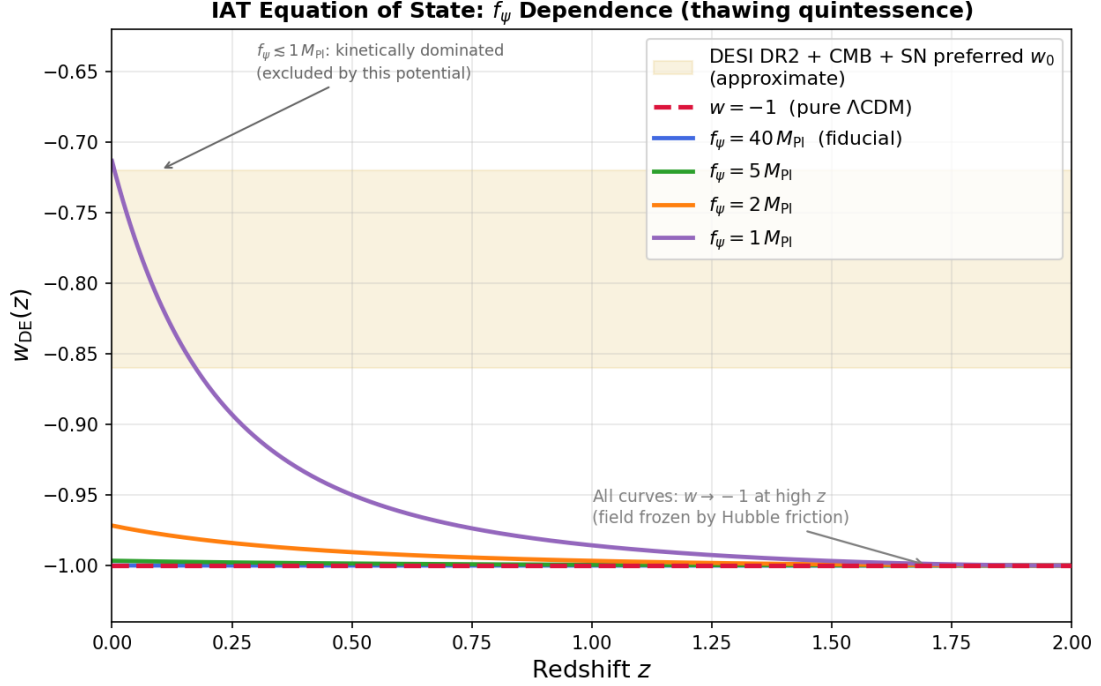


Figure 3: **IAT equation of state as a function of f_ψ .** All curves exhibit thawing quintessence behavior: the field is frozen by Hubble friction at high redshift ($w \rightarrow -1$, $z \gtrsim 1.5$) and begins rolling as H decreases, producing a present-day deviation that grows with decreasing f_ψ . The fiducial value $f_\psi = 40 M_{\text{Pl}}$ is indistinguishable from pure ΛCDM . Values $f_\psi \sim 1 M_{\text{Pl}}$ place w_0 within the range preferred by the DESI DR2 combined analysis (shaded band). Values $f_\psi \lesssim 0.5 M_{\text{Pl}}$ produce kinetically dominated solutions inconsistent with late-time acceleration, providing a natural lower bound on the decay constant.

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